

LMNA p.H222P mutation causes contractile dysfunction via impaired mitochondrial calcium uptake in human cardiac laminopathy

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Aims

Mutations in the *LMNA* gene, which encodes lamin A/C, cause a variety of diseases known as laminopathies. Some mutations are particularly associated with the occurrence of dilated cardiomyopathy and heart failure, but the genotype–phenotype relationship and underlying mechanisms are unclear.

Methods and results

Induced pluripotent stem cells (hiPSCs) from a patient carrying a *LMNA* point mutation (c.665A>C, p.His222Pro) and a CRISPR/Cas9 corrected isogenic control hiPSC clones were differentiated into cardiomyocytes (hiPSC-CMs), with no difference in the differentiation yield and in sarcomere organization between the two cell lines. However, 3D cardiac organoids generated with *LMNA* p.H222P hiPSC-CMs showed an impaired contractility compared to control organoids. Calcium transient recordings in *LMNA* p.H222P mutant cardiomyocytes showed a significantly higher calcium transient amplitude with a significantly slower calcium reuptake. Transcriptomic analyses suggested a global mitochondrial dysfunction and in particular an impaired mitochondrial calcium uptake with a significantly decreased expression of the mitochondrial calcium uniporter (MCU). This decrease in MCU expression was confirmed by western blot and was accompanied by an increased MICU1:MCU ratio, as well as an increased PDH Ser232 and PDH Ser300 phosphorylation, indicating an altered mitochondrial calcium uptake in the *LMNA* mutant hiPSC-CMs. Consistently, lower mitochondrial respiration and ATP levels were found in *LMNA* p.H222P hiPSC-CMs as compared to isogenic controls. Strikingly, treatment with the MCU activator amorolfine restored mitochondrial calcium uptake and improved contractility in *LMNA* mutant hiPSC-CMs.

Conclusion

Our results establish a direct mechanistic link between nuclear envelope dysfunction and impaired mitochondrial function and highlight the MCU complex as a potential therapeutic target in *LMNA*-related cardiomyopathy. More broadly, this work provides a paradigm for connecting gene-specific nuclear defects to mitochondrial dysfunction in inherited cardiomyopathies.

Keywords

Heart failure • Laminopathy • iPS-derived cardiomyocytes • Calcium • Mitochondria

1. Introduction

Lamins are the major structural components of the nuclear lamina, a mesh-like structure located at the inner nuclear membrane.¹ Mutations in *LMNA*, which encodes the A-type lamins (i.e. lamin

A/C), cause a variety of diseases, called laminopathies. The majority of *LMNA* mutations are associated with muscular disorders,² which often include the cardiac muscle, resulting in *LMNA* cardiomyopathy. Lamin A/C-related cardiomyopathy is one of the most common causes of familial dilated cardiomyopathy (DCM) and is

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characterized by a progression to heart failure in the early adulthood, associated with life-threatening cardiac arrhythmias, conduction defects, and a high risk of sudden death.^{3,4} It is estimated that mutations in *LMNA* account for 5–8% of familial DCM, and DCM patients carrying *LMNA* mutations are now recognized as high-risk patients requiring specific guideline-recommended management including implantable cardioverter defibrillator (ICD) implantation and early access to heart transplantation.^{4,5} However, the pathological mechanisms underlying the *LMNA*-associated cardiac phenotype are not well understood, thereby limiting the development of targeted therapeutic approaches.

Lamins support the mechanical integrity of the nucleus, are involved in chromatin organization, and interact with several other nuclear membrane proteins to modulate the cellular mechanotransduction and activation of signalling pathways.¹ Thus, *LMNA* mutations affect the nuclear envelope and have been shown to affect gene expression by inducing chromatin remodelling or the mechanical response to stress by affecting the LINC (linker of nucleoskeleton and cytoskeleton) complex at the nuclear surface. For example, the *LMNA* p.K219T mutation has been shown to induce an epigenetic inhibition of *SCN5A*, the gene encoding the sodium channel, resulting in decreased sodium current density, slower conduction velocity, and conduction abnormalities.⁶ In addition, *LMNA* mutations can alter the organization of actin and microtubule networks and the modulation of α -tubulin acetylation levels improved the cardiac function of *Lmna*^{p.H222P/H222P} mice.⁷ Additional functions are likely to be affected by *LMNA* mutations, supporting the transition to heart failure. An imbalance between oxidation and antioxidant response has been shown in *Lmna*^{p.H222P/H222P} mice, with a decrease in glutathione peroxidase levels that was reversed by *N*-acetylcysteine administration.⁸ In addition, a decrease in NAD^+ levels was also found in *Lmna*^{p.H222P/H222P} mice, and reversed by supplementing animals with the NAD^+ precursor nicotinamide riboside (NR).⁹ Metabolic abnormalities may contribute to the pathogenesis of heart failure, but whether they are also important contributors to *LMNA*-related cardiomyopathy and heart failure is not firmly established.

In this study, we focused on the *LMNA* c.665A>C p.His222Pro mutation causing the Emery-Dreifuss muscular dystrophy (EDMD), which is characterized by joint contractures and muscle atrophy and weakness associated with DCM.¹⁰ We aimed to develop a human cardiac model of the *LMNA*-related cardiomyopathy using mutant and CRISPR/Cas9 corrected human-induced pluripotent stem cells (hiPSCs) to elucidate the pathological abnormalities leading to the development of contractile dysfunction.

2. Methods

Details on the procedures are provided in the online-only data supplement. The data and material that support the findings of this study are available from the corresponding author upon reasonable request.

2.1 Induced pluripotent stem cell lines

The hiPSC line heterozygous for the NM_170707.4(*LMNA*): c.665A>C (p.His222Pro) mutation was derived by Sendai virus infection from PBMCs of a 23-year-old male patient¹⁰ and is referred to as the *LMNA* p.H222P line. The non-isogenic SKIPSC-31.3 hiPSC line, derived from human dermal fibroblasts from a healthy 45-year-old control subject, as previously reported,^{11,12} was used as a reference for the functional assessments of the cardiomyocytes. Patients gave their written consent for iPSC generation and further biological experiments as performed in this study. This study conforms to the principles outlined in the Declaration of Helsinki.

2.2 Cardiomyocytes differentiation and culture

Mutant and corrected hiPSCs were differentiated when they reached 90% confluence, according to a protocol adapted from Garg et al.¹³ and previously described in other recent studies,¹² which includes a small-molecule-driven mesoderm induction [using IWR-1 (Sigma)], and two rounds of glucose starvation to increase the percentage of cardiomyocytes obtained. After Day 18, the cells were cultured in RPMI-B27 with insulin and the medium was changed every 2 days.

2.3 3D Cardiac rings generation and recording

3D cardiac rings were generated as previously described.¹⁴ Briefly, the tissues are generated on a commercial 96-well plate containing a moulded hydrogel in each well that induces seeded cells to self-organize into up to 21 ring-shaped tissues per well (SmartHeart® assay 4DCell, Montreuil, France). At D22 of differentiation, cardiomyocytes from *LMNA* p.H222P mutant and corrected cell lines were dissociated and mixed with dissociated healthy human dermal fibroblasts at a hiPSC-CM:fibroblast ratio of 3:1, to produce mutant or corrected cardiac tissues, respectively. At 10 days after their generation, the tissues were recorded in bright-field imaging with a high-speed CCD camera (PLD672MU, Pixelink, Ottawa, Canada) mounted on a microscope (Primovert, Zeiss, Oberkochen, Germany), at a 10 $\times$ magnification. In amorolfine experiments, *LMNA* p.H222P cardiac tissues were first recorded at baseline, and then cultured in the presence of amorolfine 10 μM (Sigma, St Louis, Missouri, USA) or DMSO (0.1% v/v) for 7 days. Automated video analysis was performed using a Matlab script, which allowed the recovery of contractility parameters such as the contraction frequency, the contraction stress, and the maximum contraction and relaxation speed.

2.4 Statistical analysis

All numerical results are expressed as mean $\pm$ standard deviation (SD) of three independent experiments. The number of samples (*n*) used in each experiment is recorded in the text and figure legends. Quantitative variables were analysed with the Mann-Whitney *U* test or *t*-test as appropriate for the comparison of data obtained in the two studied groups, *LMNA* p.H222P vs. the isogenic control (Ctrl-Iso1). Data from the unrelated healthy hiPSC line are reported as a reference but are not statistically compared to the other two groups. *P*-values < 0.05 were considered significant for all statistical tests. Statistical analyses were performed with GraphPad Prism software.

3. Results

3.1 Generation and characterization of *LMNA* p.H222P mutant and corrected hiPSCs-derived cardiomyocytes

We used a hiPSC line generated from a patient heterozygous for the *LMNA* c.665A>C mutation. This missense variant leads to a change in the amino acid sequence (p.His222Pro) associated with an autosomal dominant Emery-Dreifuss muscular dystrophy (EDMD2) and *LMNA*-related cardiomyopathy as described previously.¹⁰ To obtain an isogenic corrected clone, we used CRISPR/Cas9 to correct the *LMNA* c.665A>C mutation. A mutation-specific gRNA was designed. We found that the presence of the C variant on position 665 of the *LMNA* gene induces the generation of a PAM sequence that can be specifically recognized by the CRISPR/SpCas9 system (Figure 1A). We then designed an optimized donor matrix to correct the DNA through homologous directed repair, with silent mutations to the recombination matrix to prevent unwanted re-cutting of the corrected DNA as well as gRNA homology on the donor matrix, in line with previous

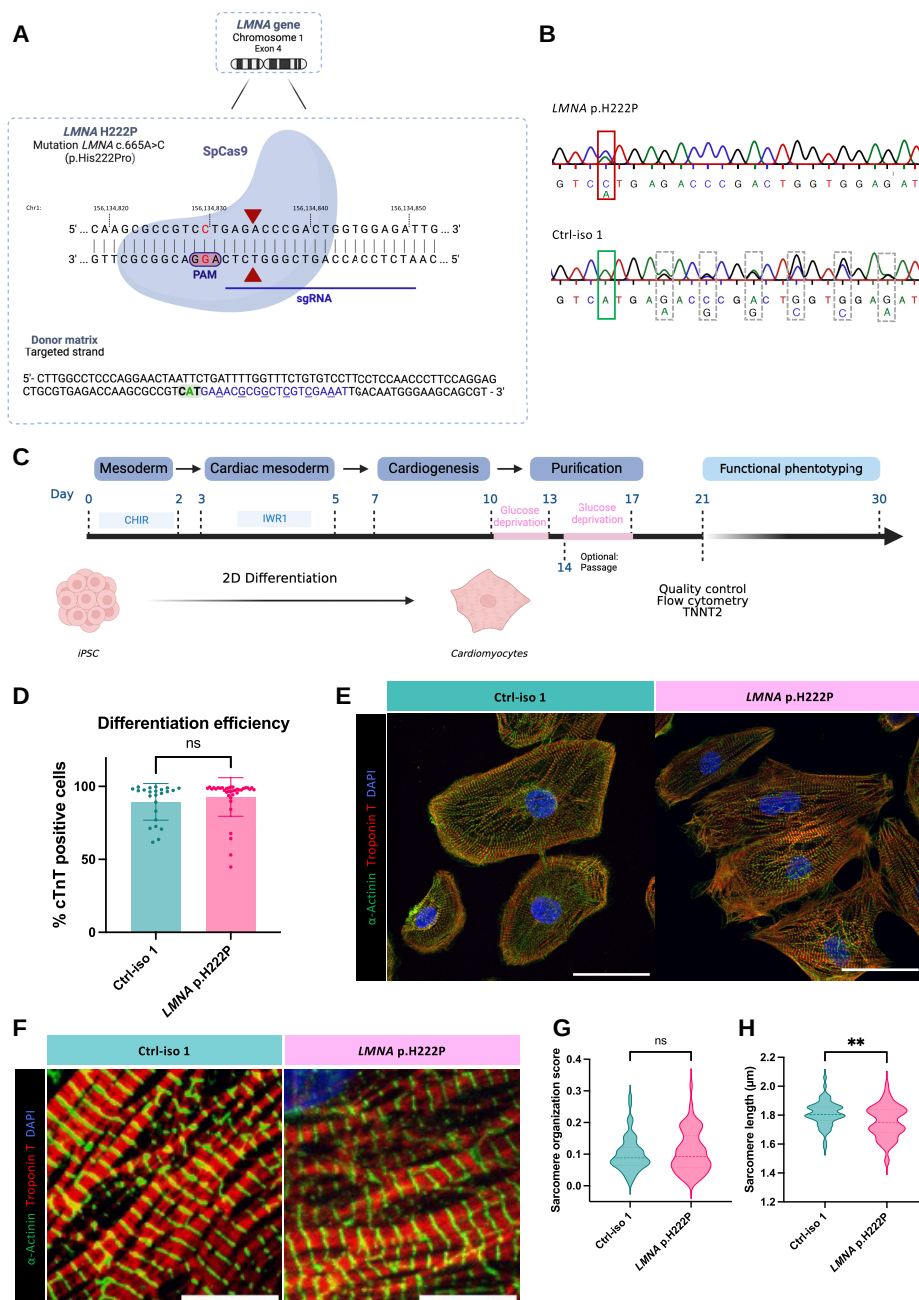


Figure 1 Correction of the LMNA c.665A>C (p.His222Pro) mutation in heterozygous hiPSCs and differentiation in cardiomyocytes. (A) CRISPR-Cas9 strategy used for the correction of the LMNA: c.665A>C (p.His222Pro) mutation (nucleotide in red). The gRNA sequence used to cut the DNA in the vicinity of the mutation is underlined in blue on the 3' strand. The PAM sequence (AGG) on the 3' strand is mutation-specific. The donor matrix was designed on the targeted strand (36 nucleotides on the PAM-distal side, 91 nucleotides on the PAM-proximal side) and displays the corrected sequence (corrected nucleotide in green) and six synonymous SNPs (underlined). The part of the donor matrix sequence, which is complementary to the gRNA is in blue. (B) Sequences of the LMNA p.H222P mutant hiPSC clone and of the corrected clone Ctrl-iso 1. The heterozygous c665A>C mutation (red frame in the mutant clone) is not found in the corrected clone (green frame), which is heterozygous for the synonymous SNPs inserted in the donor matrix (framed in grey). (C) Protocol of differentiation used to obtain cardiomyocytes from hiPSCs. (D) Differentiation efficiency expressed as the percentage of troponin T-positive cells measured by flow cytometry 21 days after the beginning of differentiation in the LMNA p.H222P mutant vs. Ctrl-iso 1. Mean $\pm$ SD, $n = 26$ and 37 differentiations, respectively, for Ctrl-iso 1 and LMNA p.H222P. Mann-Whitney test between LMNA p.H222P and Ctrl-iso 1. ns: $P > 0.05$. (E) Representative staining of alpha-actinin (green), troponin T (red), and DAPI (blue) in Ctrl-iso 1 and LMNA p.H222P hiPSC-CMs, 30 days after the beginning of differentiation. Scale bar $50 \mu\text{m}$. (F) Pictures of sarcomeres immunostaining (alpha-actinin in green, troponin T in red) of Ctrl-iso 1 and LMNA p.H222P hiPSC-CMs. Scale bar $10 \mu\text{m}$. (G, H). Sarcomere organization score (G) and sarcomere length (H) determined with a Matlab code for Ctrl-iso 1 and LMNA p.H222P. $n = 57$ and 68 cells, respectively, for Ctrl-iso 1 and LMNA p.H222P, for two differentiations. Mann-Whitney test. ** $P < 0.01$; ns: $P > 0.05$. Created in BioRender. Hulot, J. (2026) <https://BioRender.com/lbbyvb3>.

reports¹⁵ (Figure 1A and Supplementary material online, Table S1). We used the ESEfinder software^{16,17} to ensure that the synonymous SNPs would not induce any major modification of the exonic splicing enhancers in the vicinity of the mutation. The gRNA sequence was inserted into a plasmid containing the sequence of SpCas9 and nucleofected with the donor matrix. The electroporated clones were isolated and sequenced, and a corrected clone showing correction of the *LMNA* c.665A>C mutation was selected (Figure 1B) and referred to as Ctrl-iso 1. We excluded the presence of off-target events by sequencing the most likely predicted off-target sites, in the corrected and mutant clones (see Supplementary material online, Figure S1). The pluripotency of Ctrl-iso 1 hiPSCs was verified by immunofluorescence and quantitative PCR of pluripotency genes (*Nanog*, *Sox2*, *Oct4*, and *TERT*) (see Supplementary material online, Figure S2A and B). No karyotype abnormality was observed in both hiPSC clones (see Supplementary material online, Figure S2C). We also verified that the correction plasmid sequence was not integrated into the DNA of the corrected clone (see Supplementary material online, Figure S2D).

Both *LMNA* p.H222P mutant and isogenic control cell lines can be differentiated into cardiomyocytes with a similar efficiency, as shown by the quantification of TNNT2-positive cells by flow cytometry (Figure 1C and D). Immunostaining for alpha-actinin and troponin T at Day 30 of differentiation shows well-formed sarcomeres in all cell lines, which are characteristics of cardiomyocytes (Figure 1E). To further analyse the contractile apparatus of the two cell lines, we used a Matlab script to quantify the sarcomere organization and the sarcomere lengths. We found no difference in sarcomere organization score in the *LMNA* mutant and corrected cell lines (Figure 1F and G). Sarcomere length was slightly but significantly reduced in *LMNA* p.H222P hiPSC-CMs compared to isogenic control cells, with a trend for a bimodal distribution in the *LMNA* p.H222P hiPSC-CMs ($1.755 \pm 0.111 \mu\text{m}$ vs. $1.811 \pm 0.086 \mu\text{m}$, $P = 0.0049$, Figure 1H). The absolute difference in sarcomere length was minimal ($\sim 0.05 \mu\text{m}$), suggesting limited structural impact. Therefore, differentiation of mutant and corrected hiPSC lines results in apparently well-formed cardiomyocytes.

3.2 *LMNA* mutant hiPSC-derived cardiomyocytes show nuclear abnormalities

A Matlab script was used to quantify additional morphological features of hiPSC-derived cardiomyocytes. We also used an unrelated hiPSC derived from a healthy donor (referred to as WT) as another reference. No difference in the cell area was observed between *LMNA* p.H222P and Ctrl-iso 1 cell lines (Figure 2A and B). These parameters were comparable to those of the non-isogenic WT hiPSC-CMs. Since the *LMNA* mutation affects the nuclear membrane, we then compared different nuclear morphology descriptors. We found that the nuclei of *LMNA* p.H222P mutant hiPSC-CMs were significantly larger but less elongated compared to isogenic corrected hiPSC-CMs in 2D cell monolayers (Figure 2A, C, and D). Importantly, lamin A/C expression was similar in cardiomyocytes derived from mutant and corrected cell lines (Figure 2E–G) and similar to the non-isogenic WT hiPSC-CMs.

3.3 Engineered cardiac tissues generated with *LMNA* p.H222P mutant hiPSC-CMs display contractile dysfunction

We then evaluated the functional characteristics of *LMNA* mutant and corrected hiPSC-CMs. Since the function is influenced by the 3D environment, including the extracellular matrix and the multicellular interactions, we generated engineered cardiac tissues (hECT) using the *LMNA* mutant, corrected, and unrelated healthy hiPSC-CMs, respectively. Briefly, hiPSC-CMs were mixed with human dermal fibroblasts at Day 22 of CM differentiation and seeded

in an assay to generate self-assembled 3D cardiac rings (Figure 3A), as previously described.¹⁴ The rings were imaged 10 days later with a high-speed camera, and the contractile properties of the cardiac tissues were assessed by tracking the area of the central pillar over time on the bright-field videos of the rings. We first observed that both *LMNA* p.H222P and Ctrl-iso 1 engineered cardiac tissues were well formed and had a similar morphology and showed contractile activity without arrhythmic events (Figure 3B and C and Supplementary material online, Movies S1 and S2). However, the engineered cardiac tissues generated with the *LMNA* p.H222P mutant hiPSC-CMs exhibited an impaired contractility with significantly lower contraction frequency, lower contraction stress, and slower maximum contraction and relaxation velocities (Figure 3C), as compared to the cardiac tissues generated with the corrected or the WT hiPSC-CMs. Conversely, the *LMNA* corrected hiPSC-CMs showed a restored contractility, similar to that measured in the non-isogenic WT control (Figure 3C). Consistent with our results showing similar sarcomeric organization between the cell lines (Figure 1E–G), immunofluorescence staining for troponin T showed that the tissues had a similar and well-formed internal structure (Figure 3D). In addition, we analysed the nuclear morphology of hiPSC-CMs in the engineered cardiac tissues and found significant differences with smaller and more elongated nuclei in tissues generated with the *LMNA* p.H222P mutant hiPSC-CMs (Figure 3E). Taken together, these results indicate that the *LMNA* p.H222P mutation affects the structure of the nuclear envelope but does not impair the ability to generate cardiomyocytes in a 2D monolayer. However, the *LMNA* p.H222P mutation is associated with a significant reduction in contractility and relaxation as demonstrated in engineered cardiac tissues and induces a significant effect on nuclear morphology in such a 3D environment.

3.4 *LMNA* p.H222P hiPSC-CMs present abnormal calcium transients, reversed by the correction of the mutation

Cardiomyocyte contractility is closely related to intracellular calcium activity.¹⁸ Therefore, we evaluated calcium transients in our *LMNA* mutant and corrected hiPSC-CMs at 30 days of differentiation by live imaging with the fluorescent dye Fluo4-AM (Figure 4A). HiPSC-CMs were efficiently paced at 1 Hz, with no significant difference between the different cell lines (Figure 4B). Of note, we did not detect any arrhythmic events on the *LMNA* mutant cells. To remove one source of variability, only cells with a contraction frequency between 0.9 and 1.1 Hz were analysed. Therefore, we compared cells in the same range of beating frequency and compared their calcium traces (Figure 4C). Strikingly, we observed a significantly increased calcium transient amplitude in *LMNA* mutant hiPSC-CMs, with a marked increase in the area under the $\Delta F/F_0$ curve for *LMNA* p.H222P mutant cardiomyocytes, a signal that was reversed in Ctrl-iso 1. In addition, *LMNA* p.H222P cardiomyocytes exhibit a significant decrease in maximum calcium decay speed, whereas the rate of rise of the calcium transient was not affected by the mutation (Figure 4D). We next assessed the SR Ca^{2+} load and cytosolic Ca^{2+} extrusion using rapid caffeine application to deplete intracellular Ca^{2+} stores. The amplitude of caffeine-induced Ca^{2+} release was unchanged between groups (Figure 4E and F), suggesting preserved SR Ca^{2+} content in *LMNA* p.H222P hiPSC-CMs. In contrast, the area under the curve (AUC) of Ca^{2+} transient after caffeine was significantly increased in *LMNA* mutant cells (Figure 4F), reflecting prolonged cytosolic Ca^{2+} exposure. These findings suggest altered Ca^{2+} buffering or clearance dynamics rather than impaired SR Ca^{2+} load in *LMNA* p.H222P hiPSC-CMs. Therefore, we first analysed the mRNA expression of key genes involved in calcium handling (*CACNA1C*, *RYR2*, *SERCA2A*, *PLN*, *CASQ2*) but found no significant difference between *LMNA*

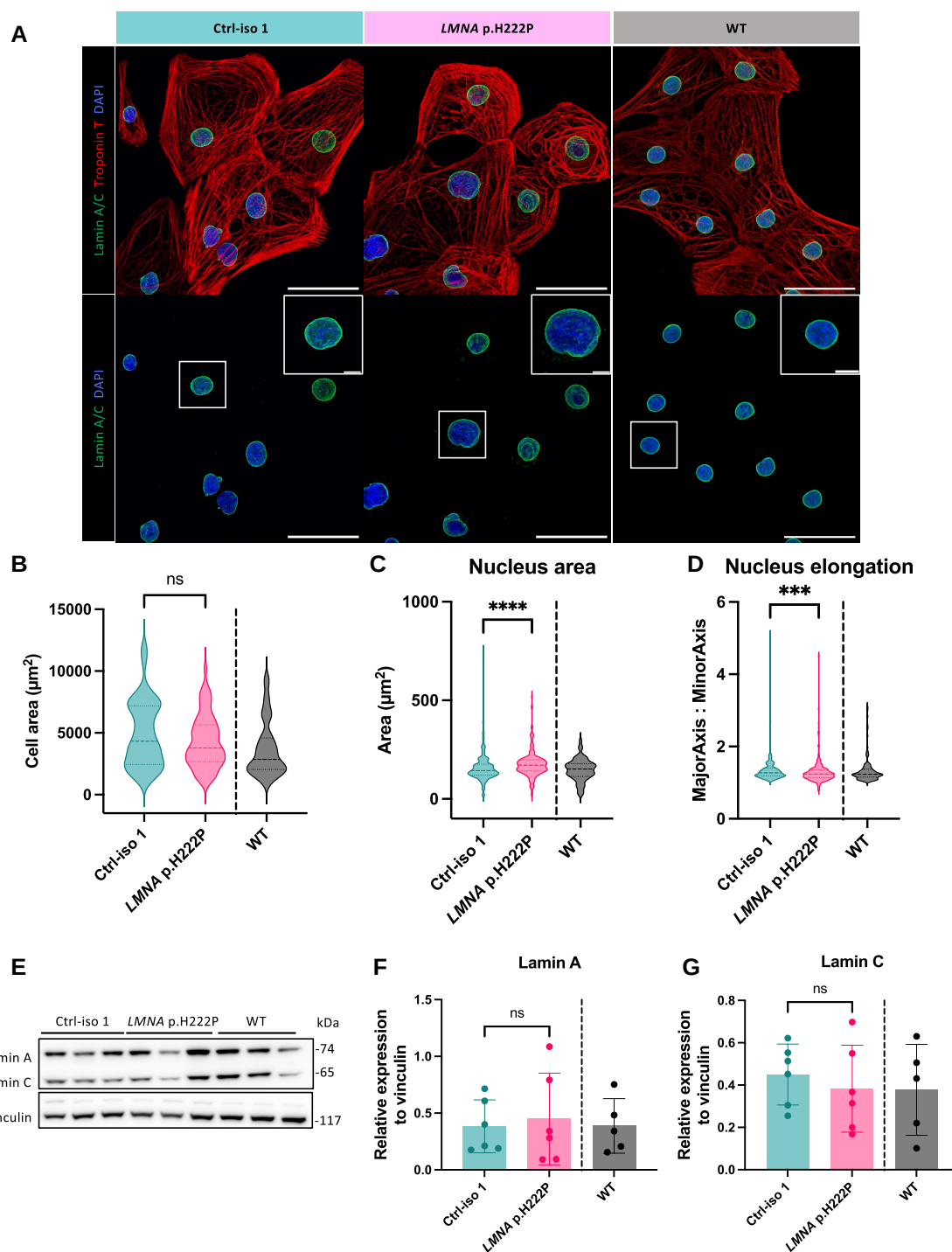


Figure 2 Cardiomyocytes morphological characterization. (A) Immunostaining of lamin A/C (green), troponin T (red), and DAPI (blue) in LMNA p.H222P and Ctrl-iso 1 hiPSC-CMs 30 days after the beginning of differentiation. Merge is shown in the upper panel and lower panel shows immunostaining of lamin A/C (green) and DAPI (blue). The insert represents a zoomed view of a nucleus. Global scale bar 50 μm , insert scale bar 10 μm . (B) Quantification of mean cell area in μm^2 measured on immunostainings for the Ctrl-iso 1, LMNA p.H222P, and non-isogenic WT hiPSC-CMs. $n = 57, 68$, and 20 cells, respectively, for Ctrl-iso 1, LMNA p.H222P and non-isogenic WT hiPSC-CMs for two differentiations. Mann-Whitney test between LMNA p.H222P and Ctrl-iso 1. ns: $P > 0.05$. (C) Nucleus area and (D) elongation quantifications in mutant and corrected hiPSC-CMs, calculated with a bioinformatic routine. $n = 223, 227$, and 98 nuclei, respectively, for Ctrl-iso 1, LMNA p.H222P, and non-isogenic WT hiPSC-CMs for two differentiations. Mann-Whitney test between Ctrl iso 1 and LMNA p.H222P. *** $P < 0.001$; **** $P < 0.0001$. (E) Western blot of lamin A/C and vinculin on Ctrl-iso 1, LMNA p.H222P and non-isogenic WT hiPSC-CMs protein extracts. $n = 3$ differentiations for each cell line. (F, G). Quantification of lamin A (F) and lamin C (G) expression in Ctrl-iso 1, LMNA p.H222P, and non-isogenic WT hiPSC-CMs, expressed as relative to vinculin expression for each sample. Mean $\pm$ SD, $n = 3$ differentiations for each cell line. Mann-Whitney test between LMNA p.H222P and Ctrl-iso 1.

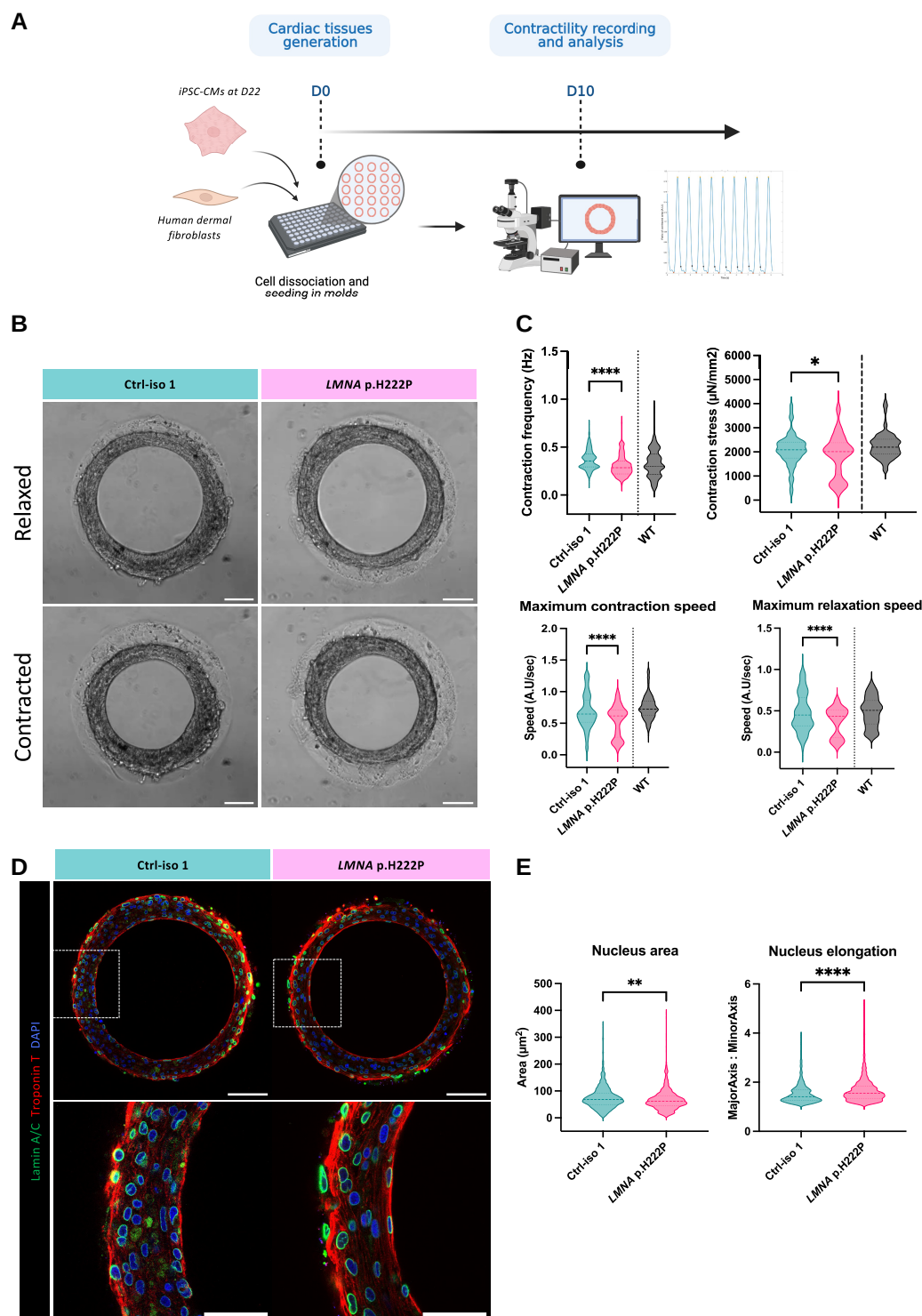


Figure 3 Tissue contractility. (A) Timeline for the tissue generation and the video recordings. (B) Bright-field images of representative tissues generated with LMNA p.H222P or Ctrl-iso 1 cardiomyocytes, in their most relaxed (upper picture) or contracted (lower picture) state. Scale bar 100 μm . (C) Contractile parameters of LMNA p.H222P and Ctrl-iso 1 tissues, compared to a non-isogenic control. $n = 164, 152$, and 183 tissues, respectively, for Ctrl-iso 1, LMNA p.H222P, and non-isogenic WT. Mann-Whitney test between LMNA p.H222P and Ctrl-iso 1. **** $P < 0.0001$; * $P < 0.05$. (D) Troponin (red), lamin A/C (green), and DAPI (blue) staining of Ctrl-iso 1 and LMNA p.H222P cardiac tissues. Scale bars 100 μm . (E) Nucleus area and elongation quantification in mutant and corrected 3D tissues. $n = 589$ and 833 nuclei respectively for Ctrl-iso 1 and LMNA p.H222P 3D cardiac tissues. Mann-Whitney test between Ctrl-iso 1 and LMNA p.H222P. ** $P < 0.01$; **** $P < 0.0001$. Created in BioRender. Hulot, J. (2026) <https://BioRender.com/z35koug>.

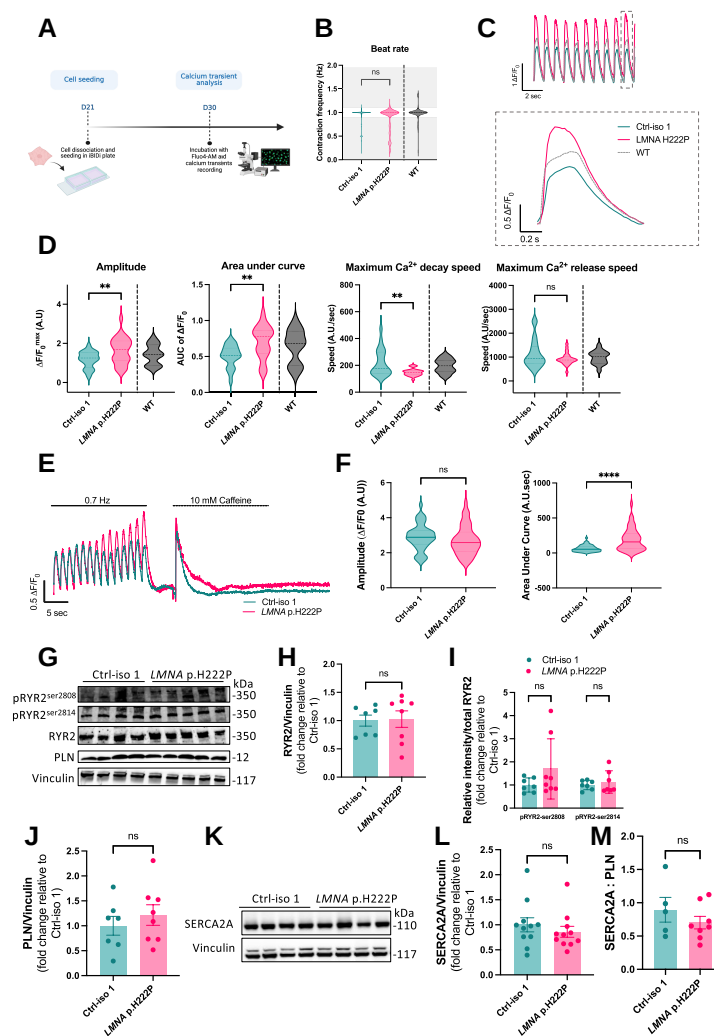


Figure 4 Calcium transient analysis. (A) Timeline for calcium transient measurements. (B) Calcium transient frequency (in Hz) of Ctrl-iso 1, LMNA p.H222P, and non-isogenic WT hiPSC-CMs. $n = 203$, 232 , and 174 cells, respectively, for Ctrl-iso 1, LMNA p.H222P, and WT non-isogenic control hiPSC-CMs. Only the hiPSC-CMs within the 0.9 – 1.1 Hz range (dotted lines) were selected for calcium transient analysis. (C) Left: Representative traces of Ctrl-iso 1, LMNA p.H222P, and non-isogenic WT hiPSC-CMs. Scale: $1\Delta F/F_0$ for y-axis, 2 s for x-axis. Right: Representative calcium transient trace for each cell line. Scale: $0.5\Delta F/F_0$ for y-axis, 0.2 s for x-axis. (D) Calcium transient parameters of hiPSC-CMs mutant or corrected for the LMNA p.H222P mutation and incubated with Fluo4-AM. Data points consist in the averaged parameter for a field: $n = 21$, 26 , and 16 fields, respectively, for Ctrl-iso 1, LMNA p.H222P, and non-isogenic WT, for three independent experiments. Mann-Whitney test between LMNA p.H222P and Ctrl-iso 1 hiPSC-CMs. **: $P < 0.01$; ns: $P > 0.05$. (E) Representative calcium transient traces recorded in Ctrl-iso 1 and LMNA p.H222P hiPSC-CMs before and after caffeine application. Cells were field-stimulated (0.7 Hz, 20 s) followed by caffeine puff (10 mM, 90 s) to estimate SR Ca^{2+} content. Scale: $0.5\Delta F/F_0$ for y-axis, 5 s for x-axis. (F) Sarcoplasmic reticulum Ca^{2+} , quantified with caffeine-induced Ca^{2+} transient amplitude (left), or area under the curve (right). Data points consist in the averaged parameter for a field: $n = 52$ and 46 fields, respectively, for Ctrl-iso 1 and LMNA p.H222P, for four independent experiments. Mann-Whitney test between LMNA p.H222P and Ctrl-iso 1 hiPSC-CMs. ****: $P < 0.0001$; ns: $P > 0.05$. (G) Representative immunoblots for total ryanodine receptor 2 (RYR2), RYR2 phosphorylated on Serine 2808 and 2814, and phospholamban (PLN). (H) Quantification of relative protein level of total RYR2 in hiPSC-CMs isolated from Ctrl-iso 1 and LMNA p.H222P, adjusted on vinculin and expressed relative to the levels observed in Ctrl-iso 1. Mean $\pm$ SD, $n = 7$ – 8 differentiations. Mann-Whitney LMNA p.H222P and Ctrl-iso 1. ns: $P > 0.05$. (I) Quantification of relative protein level of RYR2 phosphorylated on Serine 2808 and 2814 over total RYR2 in hiPSC-CMs isolated from Ctrl-iso 1 and LMNA p.H222P and expressed relative to Ctrl-iso 1. Mean $\pm$ SD, $n = 7$ – 8 differentiations. Mann-Whitney LMNA p.H222P and Ctrl-iso 1. ns: $P > 0.05$. (J) Quantification of relative protein level of PLN in hiPSC-CMs isolated from Ctrl-iso 1 and LMNA p.H222P, adjusted on vinculin and expressed relative to Ctrl-iso 1. Mean $\pm$ SD, $n = 7$ – 8 differentiations. Mann-Whitney LMNA p.H222P and Ctrl-iso 1. ns: $P > 0.05$. (K) Representative immunoblots for SERCA2A. (L) Quantification of relative protein level of SERCA2A in hiPSC-CMs isolated from Ctrl-iso 1 and LMNA p.H222P, adjusted on vinculin and expressed relative to Ctrl-iso 1. Mean $\pm$ SD, $n = 11$ differentiations. Mann-Whitney test between LMNA p.H222P and Ctrl-iso 1. ns: $P > 0.05$. (M) Proteins ratios of SERCA to PLN, calculated individually for each differentiation. Mann-Whitney test between LMNA p.H222P and Ctrl-iso 1. ns: $P > 0.05$. Created in BioRender. Hulot, J. (2026) <https://BioRender.com/z35koug>.

p.H222P vs. Ctrl-iso (see [Supplementary material online, Figure S3](#)). Consistently, protein expression levels of the canonical Ca^{2+} handling proteins (i.e. RYR, SERCA2A, and PLN) were comparable between the two groups ([Figure 4G–L](#)). The SERCA2A:PLN ratio, commonly used as an index SR Ca^{2+} reuptake capacity, was also unchanged ([Figure 4M](#)). Together, these results indicate that LMNA p.H222P hiPSC-CMs exhibit altered intracellular Ca^{2+} dynamics without overt electrical instability, despite preserved SR Ca^{2+} content and unaltered expression of major Ca^{2+} -handling proteins. Thus, the defective contractility observed in LMNA mutant cells cannot be fully explained by classical SR Ca^{2+} handling abnormalities, suggesting the involvement of alternative mechanisms regulating Ca^{2+} homeostasis.

3.5 Transcriptomic analysis highlights metabolic differences between LMNA p.H222P hiPSC-CMs and their isogenic control

In order to decipher the underlying mechanisms, transcriptomic analysis of the LMNA p.H222P mutant and Ctrl-iso 1 hiPSC-CMs was performed. By comparing gene expression in LMNA p.H222P mutant hiPSC-CMs to Ctrl-iso1 hiPSC-CMs, we found significant differences with 24 up-regulated and 14 down-regulated genes ([Figure 5A](#)). To better correlate these expression changes with the function of hiPSC-CMs, pathway analysis was performed using ReactomeGSA. Among the differentially expressed genes, many of them were zinc finger proteins, proteins involved in cell cycle and transcription/translation, up-regulation of JNK, and Notch signalling pathways, in line with previous reports in the literature for laminopathic cardiomyocytes^{19,20} (see [Supplementary material online, Figure S4A and B](#)). However, among 43 significantly altered pathways, we found that 20 of these pathways were related to genes involved in metabolism (15 pathways) or protein metabolism (five pathways) ([Figure 5B](#)). [Supplementary material online, Figure S4A and B](#) shows the significantly altered (P -value < 0.05) metabolic pathways in LMNA p.H222P mutant hiPSC-CMs compared to their isogenic control. In particular, we observed an impairment in the oxidative metabolism, including an up-regulation of the beta-oxidation of very long chain fatty acids, lipophagy, and biological oxidations. On the other hand, the pathways associated with processes leading to ATP production, such as intracellular fatty acid transport and the electron transport chain, were down-regulated.

Several genes associated with the mitochondrial function were dysregulated in LMNA p.H222P hiPSC-CMs in the transcriptomics data. We notably found a significant reduction in the expression of mitochondrial calcium uniporter (MCU) a transmembrane protein that allows the passage of calcium ions from the cytosol into the mitochondria, in the LMNA p.H222P cells ([Figure 5C](#)). Similarly, other genes important to the mitochondrial function were significantly reduced, including genes associated with mitophagy (SPATA18), cytochrome P450 electron transfer system (FDXR), and a stabilizing component of the Complex I of the mitochondrial respiratory chain (NDUFAF7) ([Figure 5C](#)). Conversely, we found no change in the expression of key cardiac genes (see [Supplementary material online, Figure S5A and B](#)).

We then further interrogated a transcriptomics profiling previously performed in murine hearts from *Lmna*^{p.H222P+/-} transgenic mice vs. controls.⁷ Consistent with our results from hiPSC-CMs, GO analysis showed that metabolic pathways were significantly dysregulated in hearts from *Lmna*^{p.H222P+/-} transgenic mice, including fatty acid metabolism and TCA cycle (see [Supplementary material online, Figure S6A](#)). In line with our results focusing on genes involved in mitochondria, we also observed a significant reduction in the expression of *Mcu* and *Ndufa7* (see [Supplementary material online, Figure S6B](#)). Overall, these results

indicate a significant alteration in genes involved in mitochondrial processes.

3.6 LMNA p.H222P mutant hiPSC-CMs show an impaired mitochondrial calcium uptake

Following the transcriptomic analyses suggesting mitochondrial dysfunction in LMNA mutant hiPSC-CMs, we further examined the expression of proteins composing the mitochondrial calcium uniporter complex (MCUC) ([Figure 5D](#)). Consistent with the RNA sequencing results, western blot analyses confirmed the significant reduction in MCU subunit expression in LMNA p.H222P hiPSC-CMs ([Figure 5E and F](#)). In addition, we found a significant increase in the expression of the regulatory subunit MICU1 ([Figure 5E, F](#)), resulting in a strong increase in the MICU1:MCU (4.332 ± 3.178 vs. 1.521 ± 1.179 , $P < 0.05$, [Figure 5G](#)). MICU2 expression was decreased in LMNA p.H222P hiPSC-CMs but the MICU2:MCU was unchanged between the groups (see [Supplementary material online, Figure S7A–C](#)).

The increased MICU1:MCU has previously been associated with reduced mitochondrial calcium uptake.²¹ To test this, we measured the pyruvate dehydrogenase (PDH) phosphorylation in LMNA p.H222P and Ctrl-iso1 hiPSC-CMs. PDH phosphorylation is modulated by the mitochondrial calcium concentration as dephosphorylation of PDH is catalysed by the pyruvate dehydrogenase phosphatase 1 (PDP1), whose activity is enhanced when mitochondrial calcium levels increase ([Figure 5D](#)). We first verified that the total PDH content was unchanged ([Figures 5H, I](#)), as well as PDP1 and PDK4 expressions (see [Supplementary material online, Figure S7D and E](#)). The three phosphorylatable serines of PDH were then evaluated: Ser 232, Ser 293, and Ser 300. We found significant increases in the pPDH^{Ser232}:PDH^{total} and pPDH^{Ser300}:PDH^{total} ratios in LMNA p.H222P mutant hiPSC-CMs ([Figure 5J](#)). Moreover, the expression of the mitochondrial $\text{Na}^+/\text{Ca}^{2+}$ exchanger (NCLX) was unchanged (see [Supplementary material online, Figure S7F and G](#)). Therefore, together with the previous observations, these results highlight the remodelling of the MCUC in LMNA p.H222P hiPSC-CMs, leading to altered mitochondrial calcium uptake and calcium-dependent activation of mitochondria as a pathogenic feature that appears from the early stage of the LMNA-related cardiomyopathy.

3.7 LMNA p.H222P hiPSC-CMs show an impaired mitochondrial respiration

As PDH activity modulates the production of Acetyl-CoA, which is implicated in the TCA cycle, which produces substrates used in the electron transport chain (ETC) for ATP production, we next evaluated the mitochondrial respiration activity in the LMNA p.H222P mutant hiPSC-CMs in comparison to their isogenic control. First, we measured the mitochondrial volume by flow cytometry and did not find any difference between the corrected and mutant hiPSC-CMs ([Figure 6A and B](#)). The assessment of mitochondrial respiration activity was then carried out with a Seahorse XFp analyser. Thirty-five days after the beginning of differentiation, hiPSC-CMs were seeded on a micropatterned substrate, which was found to improve their metabolism,²² were transferred into Seahorse plates 7 days after, and were analysed with a Seahorse flux analyser at D46 ([Figure 6C](#)). hiPSC-CMs were sequentially submitted to (i) the ATP synthase inhibitor oligomycin, (ii) the respiratory uncoupler FCCP, and (iii) the respiratory chain blockers rotenone and antimycin A ([Figure 6D](#)). This allowed us to derive the basal and maximal respirations of the cell, as well as the ATP production. We found that basal and maximal respirations were significantly impaired in LMNA p.H222P cells, as well as the ATP-linked respiration ([Figure 6E](#)). The proton leak, the coupling efficiency, the spare respiratory capacity and the non-mitochondrial oxygen consumption

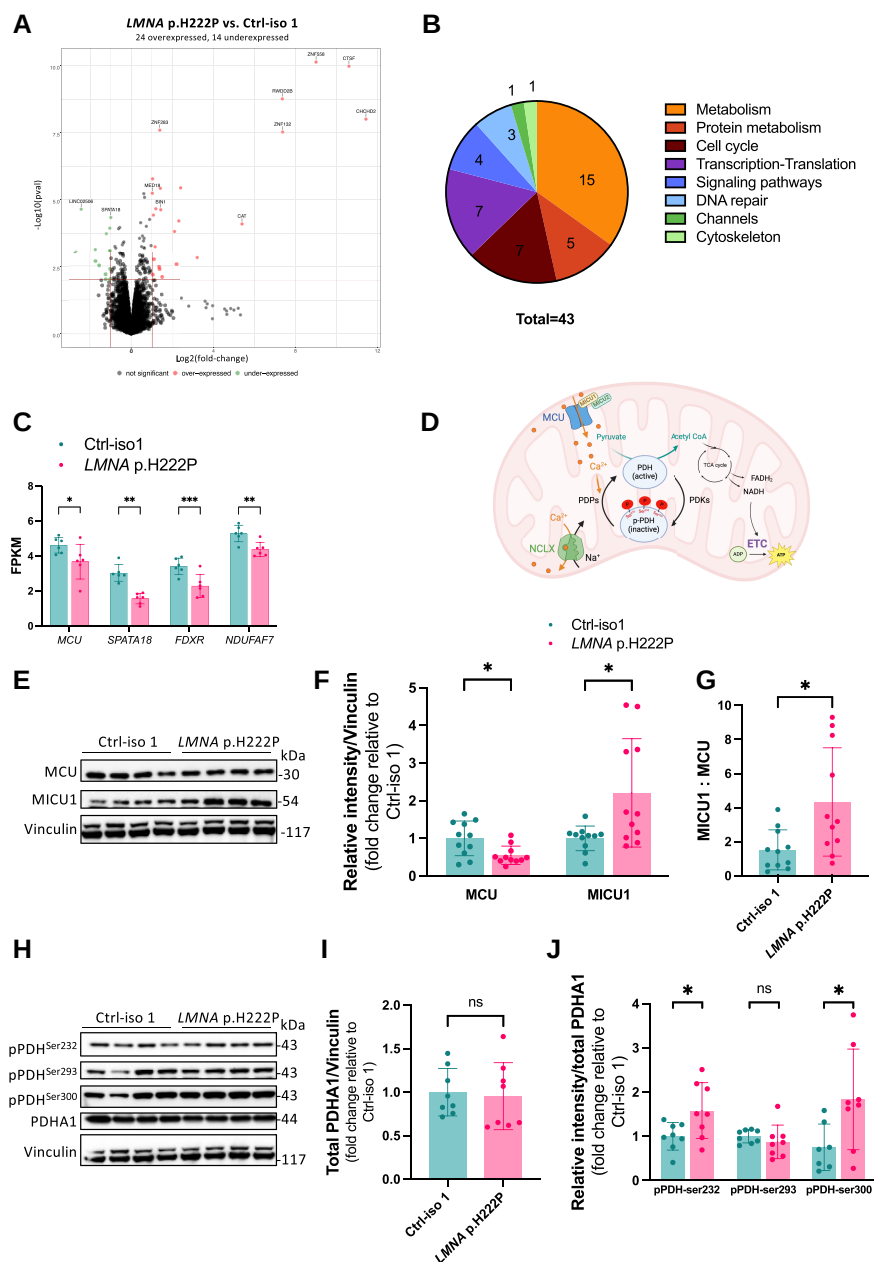


Figure 5 Impaired mitochondrial calcium handling in *LMNA* p.H222P mutant compared to Ctrl-iso 1 hiPSC-CMs. (A) Volcano plot of down-regulated (14, in green) and up-regulated (24, in red) genes in the *LMNA* p.H222P hiPSC-CMs in comparison to their corrected isogenic control Ctrl-iso 1 hiPSC-CMs. FPKM >1; fold-change >2 and P -value < 0.01. (B) Classification of significantly changed pathways (P -value < 0.05) in *LMNA* p.H222P vs. Ctrl-iso 1 hiPSC-CMs with ReactomeGSA pathway analysis. (C) RNA-seq based gene expression of genes related to the quality and efficiency of mitochondria: *MCU*, *SPATA18*, *FDXR*, and *NDUFAF7*. * P < 0.05; ** P < 0.01; *** P < 0.001. Mean $\pm$ SD, n = 6 differentiations. Mann-Whitney test between *LMNA* p.H222P and Ctrl-iso 1. (D) Scheme depicting the main actors of the mitochondrial calcium uptake and processes involved in energy production in mitochondria: MCU in blue, NCLX in green, active and inactive PDH with its phosphorylation sites in red. (E) Representative immunoblots for MCU and MICU1. (F) Quantification of relative protein level of MCU and MICU1 in hiPSC-CMs isolated from Ctrl-iso 1 and *LMNA* p.H222P. Vinculin was used as loading control and expressed relative to Ctrl-iso 1. Mean $\pm$ SD, n = 11 differentiations. Mann-Whitney test between *LMNA* p.H222P and Ctrl-iso 1. * P < 0.05. (G) Proteins ratios of MICU1 to MCU, calculated individually for each differentiation. Unpaired t -test between *LMNA* p.H222P and Ctrl-iso 1. * P < 0.05. (H) Representative immunoblots for total pyruvate dehydrogenase subunit E1 α (PDH-E1 α), PDH phosphorylated on Serine 232, 293, and 300. (I) Quantification of relative protein level of PDH-E1 α in hiPSC-CMs isolated from Ctrl-iso 1 and *LMNA* p.H222P. Vinculin was used as loading control and expressed relative to Ctrl-iso 1. Mean $\pm$ SD, n = 8 differentiations. Unpaired t -test between *LMNA* p.H222P and Ctrl-iso 1. ns: P > 0.05. (J) Quantification of relative protein level of PDH phosphorylated on Serine 232, 293, and 300 over total PDH-E1 α in hiPSC-CMs isolated from Ctrl-iso 1 and *LMNA* p.H222P and expressed relative to Ctrl-iso 1. Mean $\pm$ SD, n = 8 differentiations. Unpaired t -test between *LMNA* p.H222P and Ctrl-iso 1. * P < 0.05. Created in BioRender. Hulot, J. (2026) <https://BioRender.com/z3we7l6>.

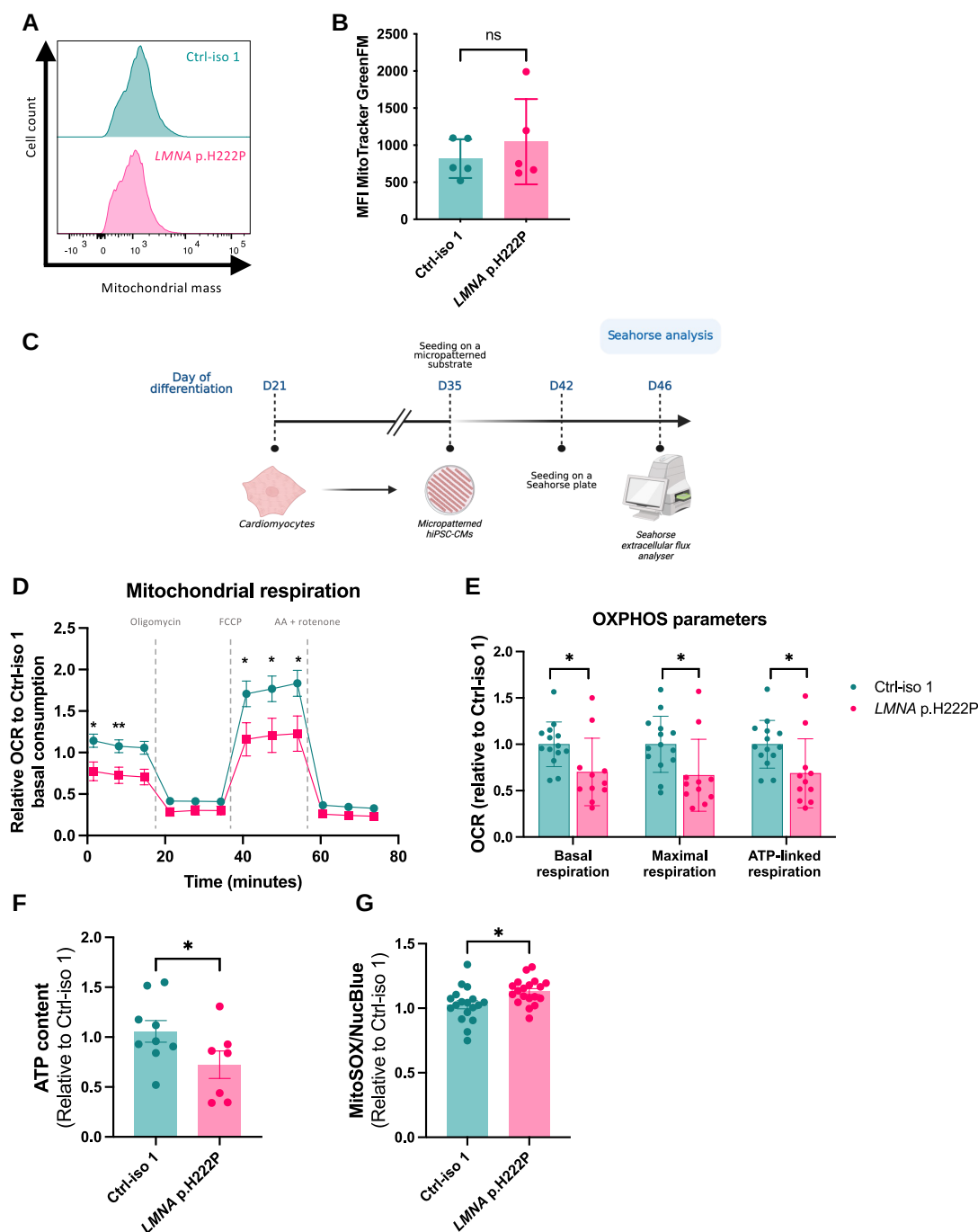


Figure 6 Impaired OXPHOS activity in LMNA p.H222P mutant compared to Ctrl-iso 1 hiPSC-CMs. (A) Representative flow cytometry images from Ctrl-iso 1 and LMNA p.H222P hiPSC-CMs stained with MitoTracker™ Green FM fluorescent dye. (B) Quantification of the Mean Fluorescence Intensity of MitoTracker™ Green FM from Ctrl-iso 1 and LMNA p.H222P hiPSC-CMs. Mean $\pm$ SD, $n = 5$ differentiations with triplicates. ns: $P > 0.05$, Mann-Whitney test. (C) Workflow of the mitochondrial respiration measurement on corrected (Ctrl-iso 1) and mutant (LMNA p.H222P) hiPSC-CMs. (D) Real-time oxygen consumption rates measurements in LMNA p.H222P and Ctrl-iso 1 hiPSC-CMs by Seahorse extracellular flux analyser. Cells were sequentially submitted to several drugs: (1) the ATP synthase inhibitor oligomycin, (2) the respiratory uncoupler FCCP, and (3) the respiratory chain blockers rotenone and antimycin A. Mean $\pm$ SD, $n = 10$ wells for LMNA p.H222P and $n = 13$ wells for Ctrl-iso 1, from three independent experiments. Results are adjusted on total protein content for each condition. (E) Comparison of the basal respiration, maximal respiration, and the ATP-linked respiration of corrected and mutant hiPSC-CMs. $n = 10$ wells for LMNA p.H222P and $n = 13$ wells for Ctrl-iso 1, from three independent experiments. Mean $\pm$ SD. Mann-Whitney test, $*P < 0.05$. (F) Measurements of ATP levels in LMNA p.H222P and Ctrl-iso 1 hiPSC-CMs. Mann-Whitney test, $*P < 0.05$. $n = 8$ from three independent experiments. (G) Measurements of ROS (with MitoSOX fluorescent probe) in LMNA p.H222P and Ctrl-iso 1 hiPSC-CMs. Unpaired t-test, $*P < 0.05$. $n = 18-19$ replicates from three independent experiments. Created in BioRender. Hulot, J. (2026) <https://BioRender.com/z35koug>.

were unchanged (see [Supplementary material online, Figure S8A–D](#)). We next quantified cellular ATP levels with a luminescence-based ATP detection assay ([Figure 6F](#)) and found that ATP content was significantly lower levels in LMNA mutant cells as compared to isogenic controls, thus confirming impaired energy production in the LMNA mutant hiPSC-CMs. Using a different assay, we also measured ADP levels and observed a significant increase in LMNA p.H222P hiPSC-CMs, accompanied by a trend towards a reduced ATP/ADP ratio (see [Supplementary material online, Figure S9A–C](#)), indicating an imbalance in cellular energy metabolism. In addition, analysis of mitochondrial superoxide production using MitoSOX fluorescence revealed a significant increase in LMNA p.H222P cells ([Figure 6G](#)), consistent with elevated mitochondrial reactive oxygen species production.

3.8 Amorolfine, a positive MCU activator, leads to functional improvement in LMNA p.H222P hiPSC-CMs

Finally, we tested whether increasing MCUC activity with pharmacological agents could reverse the functional abnormalities observed in the LMNA mutant hiPSC-CMs. It has recently been reported that the FDA-approved drug amorolfine increases MCU-dependent mitochondrial metabolism in skeletal muscle and reverses muscle atrophy.²³ We found that application of amorolfine to LMNA p.H222P hiPSC-CMs did not affect the expression of MCU ([Figure 7A and B](#)) or PDH ([Figure 7A and C](#)). However, amorolfine administration resulted in a significant decrease in the phosphorylation levels of the PDH Ser 232, Ser 293, and a trend towards a decrease for Ser 300 ([Figure 7A and D](#)), confirming that amorolfine activates MCU and ameliorates the abnormal mitochondrial calcium uptake in LMNA p.H222P hiPSC-CMs. We then tested amorolfine on cardiac tissues engineered with the LMNA p.H222P mutant hiPSC-CMs ([Figure 3A](#)). Compared to vehicle-treated tissues, we found that amorolfine led to a significant increase in contractility and relaxation parameters ([Figure 7E](#)), demonstrating that MCU activation can correct the contractility and relaxation defects observed in engineered cardiac tissues derived from LMNA p.H222P mutant hiPSC-CMs.

4. Discussion

Patients with cardiac laminopathy represent a group at high risk for developing heart failure and cardiac arrhythmias.^{3,5} However, the pathological mechanisms linking the nuclear envelope abnormalities induced by LMNA mutations to the development of a reduced cardiac muscle contractility are not well understood, limiting the development of targeted therapeutic approaches. Compared to a CRISPR/Cas9 genome-edited isogenic control, we found that cardiomyocytes derived from hiPSCs generated from a patient carrying the p.H222P LMNA mutation¹⁰ show no or minor differences in their architecture and in the sarcomere assembly, but significant differences in nuclear morphology, which is a hallmark of LMNA mutations.^{1,24} Using a recently reported 3D cardiac organoid approach,¹⁴ we were able to reproduce *in vitro* the reduced contractility found in patients with cardiac laminopathy, and this contractile abnormality was reversed by the genetic correction of the LMNA mutation and improved by the MCU activator amorolfine.

A first important finding of our work is the significant change in the MCUC stoichiometry and resulting abnormalities in mitochondrial calcium-dependent pathways in the LMNA mutant hiPSC-CMs. These changes resulted from a significant reduction in the expression of MCU, the central pore on the inner mitochondrial membrane, and a significant increase in MICU1, a regulatory subunit that exerts an inhibitory effect at low calcium concentrations. Our results indicate a significant effect of the LMNA p.H222P mutation on MCU mRNA levels, suggesting a change at the transcriptional level, as has been reported for other genes in

the context of LMNA-related disorders. The role of MCUC and mitochondrial calcium in the pathogenesis of heart failure remains controversial, as both up- and down-regulation have been shown to have deleterious effects on cardiac function. It is likely that there is a biphasic effect of mitochondrial calcium on energetics, cardiac function, and redox balance. Furthermore, the ratio between MCU and its regulatory subunit MICU1 has been found to control mitochondrial calcium uptake.²¹ An elevated MICU1:MCU ratio has been shown to exert a dual effect on mitochondrial calcium handling; it suppresses calcium uptake under low cytosolic calcium conditions while enhancing uptake when calcium concentrations are high.^{25–27} Consistent with this biphasic regulation, studies have reported that MCU activation can be either beneficial or detrimental to cardiac function, underscoring the importance of maintaining mitochondrial calcium within an optimal range.²⁸ Indeed, *Mcu* deletion protected mice from cardiac dysfunction in the context of ischaemia-reperfusion in two different studies,^{29,30} but others reported an improvement of cardiac function with a moderate overexpression of MCU in a model of heart failure in guinea pig.³¹ In addition, kaempferol, an MCU activator, improved the arrhythmic phenotypes in mice with metabolic cardiomyopathy and hiPSC-CMs from catecholaminergic ventricular tachycardic patients.^{32,33} Both excessive and insufficient mitochondrial calcium levels can trigger ROS production and impair cardiac function. Therefore, precise regulation of the MICU1:MCU ratio is critical to maintain mitochondrial calcium homeostasis within the physiological range. We propose here a model in which changes in the MICU1:MCU leads to disturbed mitochondrial calcium content, and reduced mitochondrial Ca^{2+} signalling that attenuates TCA cycle flux, compromising mitochondrial redox balance and ATP generation and leading to heart failure. The down-regulation of MCU is consistent with our analyses of calcium transients, which showed a paradoxical increase in the transient amplitude and AUC associated with a decreased calcium decay speed in the LMNA p.H222P mutant hiPSC-CMs.^{34,35} It also correlates well with the impaired mitochondrial respiration and reduced ATP production. Moreover, increased MICU1:MCU associated with a decrease in ATP production would resolve the apparent contradiction between the increase in calcium transient amplitude and the decreased contractile force.

LMNA mutations have previously been shown to have metabolic effects. Mutations leading to lipodystrophies or Hutchison–Gilford progeria syndrome (HGPS) induce metabolic disturbances, increased ROS, and decreased mitochondrial respiration with an attenuated NAMPT–NAD pathway.^{36,37} Consistently, *N*-acetyl cysteine treatment or NR supplementation reduced cardiac oxidative stress injury and ameliorated contractile dysfunction in *Lmna*^{H222P/+} mice.^{8,9} Finally, impaired mitochondrial respiration has been reported in skeletal muscle cell lines carrying LMNA p.G232E or p.R482L mutations.³⁸ Here, we further report that the mitochondrial calcium uptake pathway, which is critical for energy production and has been shown to play a role in the pathogenesis of cardiomyopathies,³⁹ is likely involved in the development of cardiac laminopathies. This is also supported by another important finding from our study, which relates to the beneficial effect of the MCU activator amorolfine. Amorolfine was recently identified as one of the most potent MCU activators in a high-throughput screen comprising 1600 US Food and Drug Administration approved drugs.²³ Amorolfine was shown to increase mitochondrial calcium uptake without impinging on cytosolic calcium transients, in different cell types, and independently of the calcium transient trigger. This suggests that this drug specifically modulates the activity of the MCU. Moreover, in agreement with the positive effect of MCU in muscle trophism,⁴⁰ amorolfine was shown to increase skeletal muscle size and reverse muscle atrophy.²³ Consistently, our results now show that amorolfine corrects the abnormal mitochondrial calcium signalling pathway in LMNA p.H222P

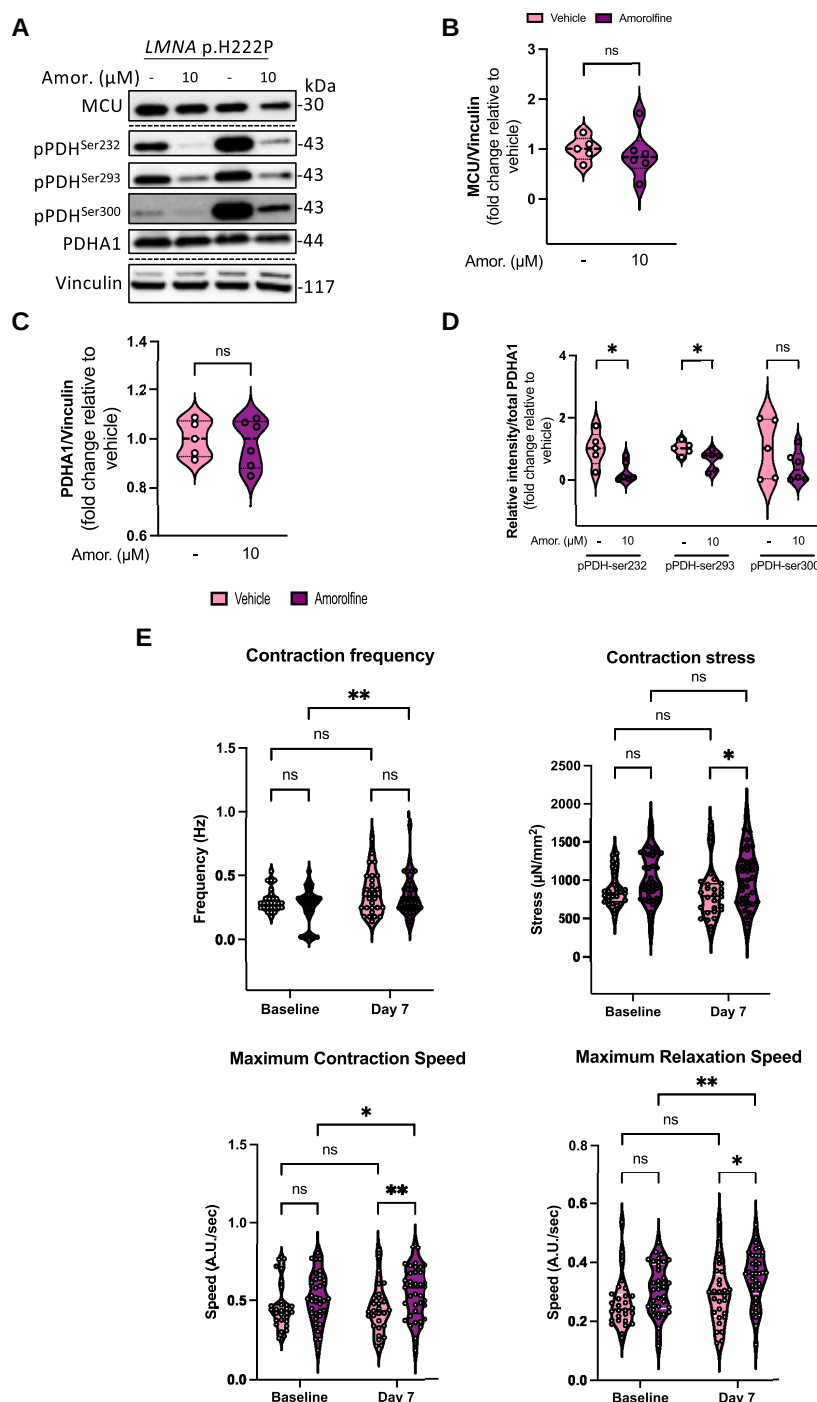


Figure 7 Amorolfine restores mitochondrial calcium handling and contractile dysfunction in LMNA p.H222P cells. (A) Representative immunoblots for MCU, total pyruvate dehydrogenase subunit E1 α (PDH-E1 α), and PDH phosphorylated on Serine 232, 293, and 300. (B) Quantification of relative protein level of MCU in LMNA p.H222P hiPSC-CMs treated with 10 μ M amorolfine or vehicle for 7 days. Vinculin was used as loading control and expressed relative to vehicle. $n = 5-6$ differentiations. Unpaired t -test between groups. ns: not significant. (C) Quantification of relative protein level of PDH-E1 α in hiPSC-CMs isolated from LMNA p.H222P treated with 10 μ M amorolfine or vehicle for 7 days. Vinculin was used as loading control and expressed relative to vehicle. $n = 5-6$ differentiations. Mann-Whitney test between groups. ns: not significant. (D) Quantification of relative protein level of PDH phosphorylated on Serine 232, 293, and 300 over total PDH-E1 α in hiPSC-CMs isolated from LMNA p.H222P treated with 10 μ M amorolfine or vehicle for 7 days and expressed relative to vehicle. $n = 5-6$ differentiations. Unpaired t -test between groups. * $P < 0.05$; ns: $P > 0.05$. (E) Contractile parameters of LMNA p.H222P treated with 10 μ M amorolfine for 7 days, compared to vehicle; $n = 37$ and 28 tissues, respectively. Two-way ANOVA of repeated measures with uncorrected Fisher's LSD. ** $P < 0.01$; * $P < 0.05$; ns: $P > 0.05$.

hiPSC-CMs, improving contractility and relaxation in LMNA mutant cardiac tissues. This translational result expands the potential applications of MCU activators as being a novel therapeutic option for patients presenting with LMNA cardiomyopathies, treating the cardiac as well as muscular symptoms.

In line with typical description in other cell types carrying LMNA mutations, the nuclear morphology was significantly changed in the 3D engineered cardiac tissues generated with the LMNA p.H222P mutant cardiomyocytes. However, these changes were different from those observed in the 2D monolayer culture conditions. Nuclei in LMNA mutant 3D tissues were smaller and more elongated than their isogenic control counterparts, consistent with the *in vivo* observations in the cardiac tissue of *Lmna*^{H222P+/-} transgenic mice.⁴¹ Similarly, our work highlights the increased deformability of the nuclei from LMNA mutant cardiomyocytes, suggesting an impaired mechano-transduction that is better revealed in 3D tissues that impose higher constraints from the surrounding cells.¹⁴ To our knowledge, our study is one of the few to investigate 3D tissue engineering using hiPSC-CMs with LMNA mutations. Two studies showed that 3D engineered skeletal muscle using LMNA mutant hiPSC-derived skeletal myotubes have an increased nucleus elongation.^{42,43} Moreover, a 3D environment was needed to reveal nuclear defects for the LMNA p.K32del and p.L35P mutations.⁴² Further investigations are now needed to better understand how the change in nuclear envelope induced by LMNA mutations can result in changes in MCU expression and activity.

Our study has several limitations. Given the rare nature of this mutation, the investigation was limited to hiPSC-CMs derived from a single patient. Replication of these findings in additional patient-derived lines carrying the same or other pathogenic LMNA variants would strengthen the generalizability of our results. In addition, while several lines of evidence support impaired mitochondrial calcium uptake, direct measurement of MCU activity remains technically challenging in hiPSC-derived cardiomyocytes.

Developing more robust tools for mitochondrial calcium imaging in these models will be an important step to further validate this mechanism.

It is also important to consider the intrinsic limitations of iPSC-derived cardiomyocytes, notably their immature state. The implementation of 3D organoid culture partially mitigated this issue by providing a more physiologically relevant geometry that enabled better detection of mutation-induced structural and functional defects. The contractility of those 3D tissues was, however, recorded under spontaneous pacing and unstressed conditions, which may lead to higher inter-tissue variability and explain the trend towards a bimodal distribution of contractility parameters observed in the LMNA p.H222P mutant cardiac tissues. Consistent with a mitochondrial dysfunction, it is likely that recording under pacing conditions would further accentuate the differences between groups, as observed with calcium transient recordings. Multiple mitochondrial functions are likely altered in the LMNA mutant hiPSC-CMs, as suggested by the deregulation of the expression of several genes involved in different mitochondrial processes. Additional experiments will be needed to further circumvent the mitochondrial abnormalities observed in patients with LMNA cardiomyopathy. Finally, although our study reveals transcriptional consequences of the LMNA mutation, it was not designed to investigate chromatin remodelling or histone modifications, which warrant focused investigation in future studies.

In conclusion, we report the existence of a novel pathogenic mechanism linking mitochondrial dysfunction with impaired MCU complex stoichiometry, altered mitochondrial calcium homeostasis, and reduced mitochondrial energy production, ultimately contributing to the contractile dysfunction caused by the Lamin A/C mutation. Overall, our study opens new therapeutic perspectives by identifying MCU activation as a potential strategy for patients with LMNA-related cardiomyopathy.

Translational perspective

Mutations in the LMNA gene cause familial cardiomyopathies, yet the molecular links to contractile dysfunction remain unclear. Using iPSC-derived cardiomyocytes carrying the LMNA p.H222P mutation, we identify impaired mitochondrial calcium uptake and disrupted MCU complex stoichiometry as key factors reducing contractility. Treatment with the MCU activator amorolfine improves contractile performance. These findings reveal mitochondrial calcium dysregulation as an unrecognized contributor in LMNA-related heart failure and highlight the MCU complex as a promising therapeutic target, offering new avenues for targeted intervention in LMNA-related cardiomyopathy.

Supplementary material

Supplementary material is available at [Cardiovascular Research](#) online.

Authors' contributions

M.S., C.J., A.R.-V., L.D., Z.G., C.P., V.R., and C.C. performed the experiments, acquired, and interpreted the data. M.S. performed the 3D cardiac organoids experiments with the help of C.J., C.C., and C.P. A.R.-V. performed bioinformatics analysis on the transcriptome data. Z.G. and M.S. performed the nuclei measurements. V.R. and M.S. performed the caffeine measurements. J.F., K.W., G.B., and A.M. reviewed the manuscript and helped in interpreting the data. J.-S.H. designed the study, supervised the experiments, analysed and interpreted the data, and wrote the manuscript with M.S. and C.J.

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Data availability

The RNA-Seq data are available in the NCBI GEO, accession number GSE304575.

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